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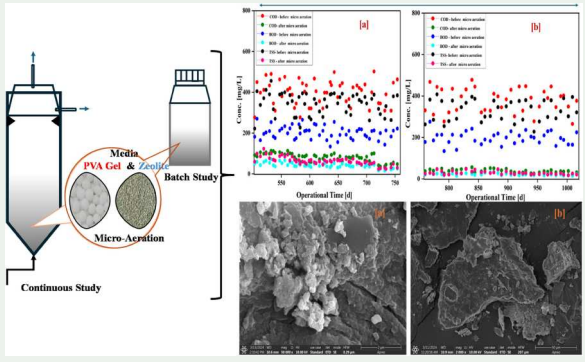
Impact of micro-aerobic conditions in UASB Reactor using media to improve the removal efficiency of organics and nutrients from wastewater

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ABSTRACT

The present study attempted to investigate the impact of the micro-aerobic condition in a conventional Up-Flow Anaerobic Sludge Blanket Reactor (UASB) using different media for the treatment of municipal wastewater. The Hydraulic Retention Time (HRT), Dissolved Oxygen (DO), intermittent and continuous aeration were the important operating conditions of this study carried out for more than 1015 days. Fourier transform infra red (FTIR) spectra and SEM analysis indicate the significant growth of the microbiota responsible for the removal of nutrients. Significant increase in the removal of the chemical oxygen demand (COD)~ 91%, biochemical oxygen demand (BOD)~ 90%, suspended solids (SS)~ 92%, and ammonical nitrogen (NH_4^+–N)~ 90%, were observed at lower DO levels. Sulphate was dominant due to sulphide oxidation during micro-aerobic conditions in almost all study phases, which resulted in an increase of up to 90% of the influent sulphates. The micro-aerobic UASB process could be a feasible option for achieving the latest treated effluent disposal norms and significantly reducing the energy demand for wastewater treatment.



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Micro-aerobic UASB system; HRT; DO; sewage

1 Introduction

The application of biological processes for the removal of organics, phosphorus, and nitrogenous compounds from municipal wastewater is extensively reported in the literature. The treatment efficiency of these processes depends on the different parameters and technology [1,2]. Usually, anaerobic systems are not as efficient as aerobic processes for the removal of organics from the wastewater [3,4], predominantly operated at small hydraulic retention times (HRT <10 h) [5].

The up-flow anaerobic sludge blanket (UASB) reactor has been used for a long time to clean wastewater in developing countries, especially tropical ones like India, Brazil, Colombia, and China [6]. Since the UASB is a robust technology, it's imperative to investigate the approaches to modify the UASB process and enhance the elimination of organics and nutrients from municipal wastewater, specifically designed for short HRTs (usually less than 8 h). Micro-aerating the anaerobic reactors, which means adding little amounts of O_2 (up to 1–2 mg O_2 L^{-1}), may be a better option in this case and may be easier to run and cost less [7], to make it easier for bacteria that make monooxygenase to break down compounds that are hard to break down, like aromatic hydrocarbons [8].

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It has already been shown that this method makes it much easier to get rid of organic matter and nutrients in an anaerobic reactor with no affecting its general performance or stability [9]. Though the earlier study only looked at a single airflow rate (1 mL min^{-1} at 27 ^{o}C and 1 atm, which is the same as 0.021 L O_2 L^{-1} feed). Because of this, the amount of O_2 used was much lower than the maximum of 1 L O_2 L^{-1} feed [10]. This means that higher microaeration flow rates could be used to make the removal of organic matter and nutrients even more effective in anaerobic reactors.

However, if there is more O_2 in the system, it may prevent the methanogenic activity, which is said to be directly related to the anaerobic biotransformation of organic compounds [11].

Thus, testing various airflow rates is important to find out the limits of this engineering approach in terms of both system stability and efficiency. This will help prove microaeration as a useful technology that can be used in large anaerobic systems to help them get rid of more organic matter and nutrients. Presently, limited studied found in the literature on micro-aeration of UASB for sewage treatment.

Hence, this study aimed to evaluate the effect of PVA-Gel and Zeolite as bio media to improve the removal of organics and nutrients from municipal wastewater. The existing UASB process was modified to a Micro-Aerobic UASB process by injecting micro-aeration in the upper zone of the reactor. Micro aeration was verified to be an effective approach to improve the biotransformation/ removal of the organics and nutrients at short HRT.

2. Materials and methods

2.1. Wastewater feed, seed sludge and biomass carrier

The sewage was brought daily from the nearby intermediate pumping station (IPS) and filled in a 1000 L capacity tank kept in the premises of the laboratory of microbial technologies for upgradation of anaerobic processes (Labmite), Department of Civil Engineering, Jamia Millia Islamia (A Central University), New Delhi. The anaerobic sludge was obtained from UASB based sewage treatment plant (STP) of capacity 111 million litres per day (MLD), Ludhiana, India.

The TS (Total Solids) and VS (Volatile Solids) of anaerobic sludge were 58 and 43 g L^{-1}, respectively. The PVA Gel and Zeolite were used simultaneously as the biomass carriers at a 30–40% reactor volume ratio. The bio media was utilised to develop the accessibility of the microorganism population to settle in and reduce the effect of outer environmental conditions.

2.2. Experimental set-up and operating conditions

2.2.1. Continuous flow studies

The continuous flow experiment was performed in a Micro-Aerobic UASB reactor fabricated from Perspex material having a working volume of 38 L at an HRT of 6 h. The reactor was fed with the municipal wastewater by a peristaltic pump (Miclins) brought daily from the nearby sewage pumping station (SPS), Batla House, New Delhi, India, and operated at a room temperature varied from 15 to 32 ^{0}C.

A micro-porous aeration device and rotâmeter were used to adjust the aeration flow rate of 100–200 mL/min to maintain DO levels between 0.5 and 2.5 mg/L. The micro-porous tube was made of polyethylene (pore size: 2.5 mm), and the aeration tube was connected to the air pump and was placed in the middle of the reactor above the sludge bed with recirculation at the bottom. The biogas produced was collected in a gas holder.

The present study was divided into two phases as summarised in Table S1. Initially, in phase 1 with intermittent aeration, the major focus of the study was on the removal of the dissimilar organics and nutrients under anaerobic conditions, which extends to explore the consequence of micro-aeration on the biotransformation/ removal of organics and nutrients at DO levels of 1.0–2.0 mg/L under operating conditions 1–3. Subsequently, under operating conditions 4-5, the adaptation of the microbiota under microaerobic conditions was evaluated. Although the stabilised concentration was achieved at each stage, the average effluent COD consistently decreased throughout all phases.

In phase 2, the effect of continuous micro-aeration on the removal/biotransformation of organics and nutrients at DO<1.0–2.0 mg/L under conditions 1–2 at HRT 30 min. Condition 3 concludes the experiment by reinforcing the oxygen effect and disproving the idea of microbial adaptability. The reactor was again operated at DO 1.0–1.5 mg/L with recycle in the aerobic zone, and in condition 4, the reactor was operated at the same DO but with recycle in the anaerobic zone. The experimental conditions were changed after verifying system stability with the average effluent COD, BOD and nutrient removal consistently decreased throughout the phase.

2.2.2. Batch studies

Experiments were also conducted in the batch reactors [100 mL serum bottles] for more than 60 days to cultivate the AnAOB (anaerobic ammonium-oxidising bacteria) and POAs (phosphorus-accumulating organisms) using different seed sludge and media. The present study was started with a synthetic feed to optimise the doses and operating conditions. The aerobic sludge was collected from the STP [see Section 2.1]. PVA Gel and Zeolite were used as the media for the attached growth of biomass. The media i.e. PVA-Gel and Zeolite were inoculated with aerobic and anaerobic sludge. The ratio of sludge to media was kept 1:1 on a weight basis. The 3 g/L of sludge along with 3 g/L of attached media were introduced in a batch reactor and kept at a constant temperature of 35 ± 2 °C. The agitation/ or mixing of sludge was done during the batch feeding of all reactors. All these batch reactors were operated in sequencing batch mode, where a hydraulic retention time of 3 d was maintained. The study was divided into eight phases [Table S2]. Each phase was planned to continue until the observation of stable performance was observed based on NH_4^+-N concentration. I stage begins with 50 mg/L supplement of NH_4^+-N followed by an increment up to 100 mg/L NH_4^+-N supplement in II stage. Subsequently, nitrite was introduced in the synthetic feed in stages III, IV, and V in the increasing order of 25, 50, and 100 mg/L, respectively. In stages VI, VII, and VIII, carbon was supplemented with synthetic wastewater to maintain ammonia: COD (chemical oxygen demand) ratios of 1:1, 1:1:5, and 1:2 to investigate the effect of carbon on enriched ammonia-oxidising bacteria.

2.3. Fourier transform infrared spectroscopy

Fourier transform infrared spectroscopy (FTIR) was conducted to assess the chemical and structural properties of sludge samples at the start and conclusion of the study (0 and 1015 days) utilising a Bruker, USA (Model; Tensor; 37) at the Central Instrumentation Facility, Jamia Millia Islamia, New Delhi. Samples were prepared in accordance with the instrumental methodology. The operational range employed was 4000–600 cm^{-1}, with a resolution of 10 cm^{-1}.

2.4. SEM analysis

Ten millilitres of sludge were put on a poly-L-lysine-coated glass slide and left to dry in the air. After that, 2.5% glutaraldehyde was used to fix the samples for 24 h at 4 °C. With phosphate buffer (pH 7.4), the fixed samples were centrifuged at 4000 rpm for one minute. The supernatant was then thrown away. For dehydrating the sample, analytical grade ethanol mixed with double-distilled water was used. The sample was then dried in a hot air oven at 110 °C. The samples were covered with 100 nm of gold and looked at with a scanning electron microscope (ZEISS ΣIGMA, Germany) at the Center for Nanoscience & Nanotechnology, Jamia Millia Islamia, New Delhi, to confirm the morphology of sludge.

2.5. Analytical procedure

All the physico-chemical parameters were analysed according to the Standard Methods of examination of water and wastewater (APHA, 18th ed, 1992).

3. Results and discussion

3.1. Batch study

3.1.1. Removal of Nitrogen and phosphorus without the use of media

Figure S1 depicts the variation of nitrogen and phosphorus levels during phases I and II of the investigation, in both aerobic and anaerobic sludge without any associated media. The results indicate that the elimination of NH_4^+-N increased from 17 to 25% over the first 9 days and continued to grow up to 86–94% at the end of the study phase I (day 30).

During phase II, it was noted that the efficacy of NH_4^+-N removal steadily decreased to 50% by day 48 and further decreased to 17–25% by the conclusion of phase II (day 60). In the aerobic sludge, the process of nitrification showed a progressive increase for the first 9 days. However, no additional nitrification was seen during phase I. However, at the same concentration of ammonium, the process of nitrification increases in the effluent at the end of phase 1. In a similar manner, the accumulation of nitrite was not observed until day 13, after which it gradually decreased until day 55 (phase II). Subsequently, as the efficacy of NH_4^+-N removal increases, the accumulation of nitrite similarly rises up. Due to partial oxidation to nitrate (denitrification), the observed trend indicates that increased NH_4^+-N removal efficiency encourages nitrite build-up. However, denitrification was not observed in phase I of the anaerobic sludge due to the absence of oxygen.

During phase II, the process of nitrification experienced a rise and reached levels ranging from 2 to 18 mg/L. Nitrite concentration was first seen on day 13 during phase I and thereafter peaked at levels ranging from 10 to 22 mg/L during phase II. At the conclusion

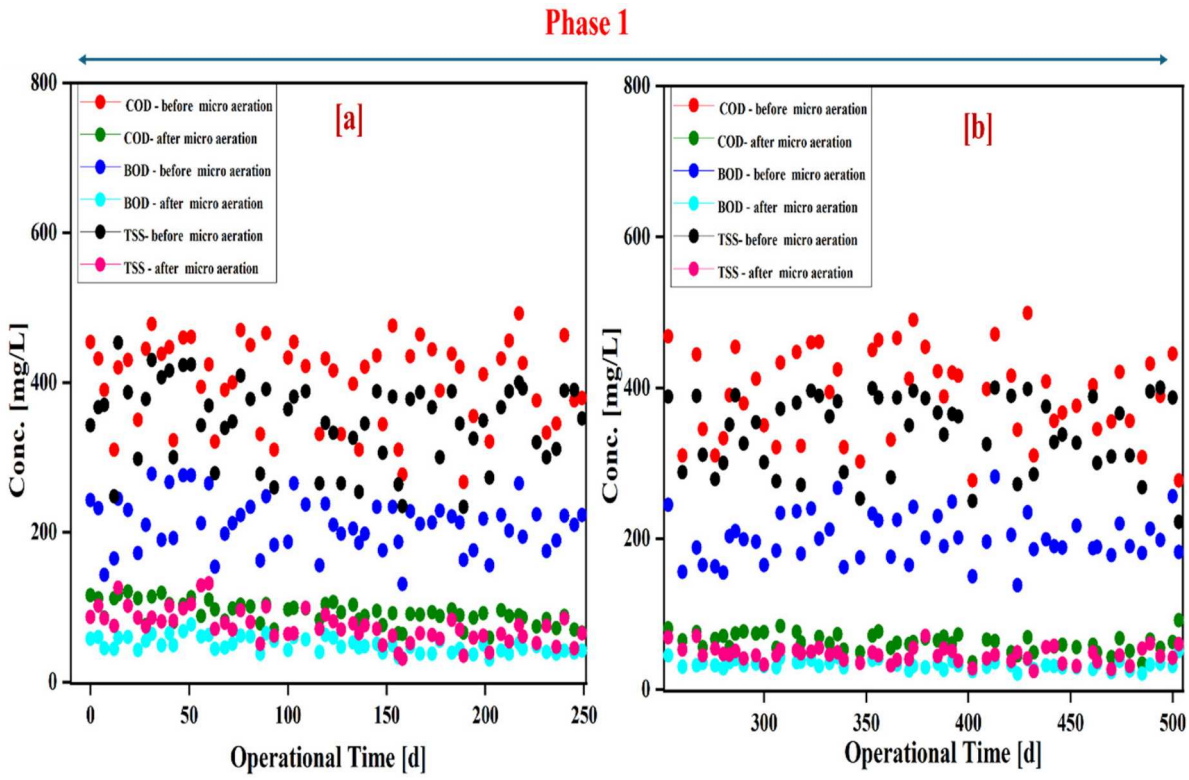


Figure 1. Treatment performance of the UASB for COD, BOD and TSS before and after micro-aeration [a] condition 1–3 and [b] condition 4–5 in phase 1.

of phase II, no nitrification nor nitrite increase was detected. The efficiency of phosphate removal was consistent and virtually identical in both aerobic and anaerobic sludge. The phosphate removal ranged from 56% to 80% in both configurations, as depicted in Figure 1.

3.1.2. Removal of nitrogen and phosphorus in sludge using PVA-gel media

The nitrogen and phosphorus concentrations in aerobic and anaerobic sludge with PVA gel were measured during phases I and II, as depicted in Figure S2. The NH_4^+-N removal efficiency in both aerobic and anaerobic sludge batch reactors was measured at 17% over the initial 9 days of the investigation. At the conclusion of phase I, which lasted 30 days, both reactors achieved an NH_4^+-N removal efficiency of 89–96%. However, the efficacy of NH_4^+-N removal rapidly decreased to 48–63% by day 48 during phase II, and further declined to 10–13% at the conclusion of phase II (day 60) in anaerobic sludge with PVA gel. In the aerobic sludge, the process of nitrification showed a progressive increase for the first 9 days.

However, no additional nitrification was seen at the end of phase I. Nevertheless, as the NH_4^+-N removal increased throughout phase II, the process of nitrification reoccurred in the effluent, resulting in a reduction to 1.4 mg/L by the end of this phase. Following that, there was no presence of nitrite accumulation until day 13. Subsequently, nitrite accumulation gradually rose until day 55, which marks the final stage of phase II. Subsequently, as the efficacy of NH_4^+-N removal declined, the accumulation of nitrite was similarly diminished. Nitrification was absent in phase I of the anaerobic sludge. During phase II, the process of nitrification experienced a rise and reached levels ranging from 2 to 18 mg/L.

Nitrite build-up was first noticed on day 13 during phase I and thereafter peaked at a maximum concentration of 23.2 mg/L during phase II. By the end of phase II, the levels of nitrification and nitrite build-up were significantly decreased to a minimal quantity of 0.6–2 mg/L. The phosphate removal efficiency was consistent and almost the same in both aerobic and anaerobic sludges. The phosphate removal efficiency ranged from 52 to 88% in both configurations, as depicted in Figure 2.

3.1.3. Removal of nitrogen and phosphorus using zeolite

Figure S3 illustrates the elimination of nitrogen and phosphorus in aerobic and anaerobic sludge using

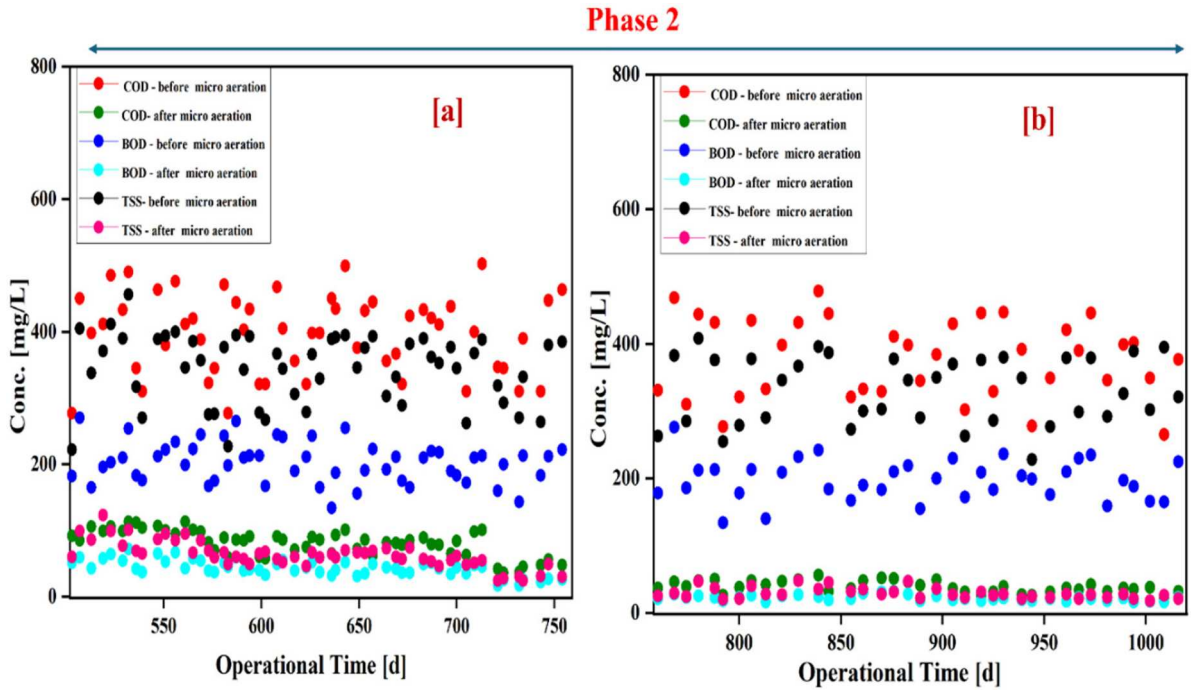


Figure 2. Treatment performance of the UASB for COD, BOD and TSS before and after micro-aeration [a] condition 1–2 and [b] condition 3–4 in phase 2.

zeolite as an attachment medium for phase I and II. The NH_4^+-N removal efficiency in both aerobic and anaerobic sludge systems reached 35–56% within the initial 9 days. At the conclusion of phase I, which lasted 30 days, both configurations achieved an NH_4^+-N removal effectiveness of 89–99%. However, the efficacy of NH_4^+-N removal rapidly decreased throughout phase II, reaching 48–63% by day 48 and ultimately dropping to 10–13% by the conclusion of phase II (day 60). The nitrification in the aerobic sludge exhibited a modest increase during the initial 9-day period. Subsequent cyclical occurrence and vanishing of nitrification were noted during phase I. Nevertheless, when the ammonium concentration increased (during phase II), the occurrence of nitrification in the effluent was continuously observed. Though the nitrate levels were consistently low, often ranging from 4 to 8 mg/L, and decreased to almost zero mg/L by the conclusion of phase II. In a similar manner, the accumulation of nitrite was not observed until day 13, after which it gradually rose until day 55 (phase II).

Subsequently, as the efficacy of NH_4^+-N removal declined, the accumulation of nitrite was similarly diminished. Comparable findings were noted with anaerobic sludge. The efficiency of phosphate removal remained consistent in both phase I and phase II, regardless of whether aerobic or anaerobic sludges were used. The phosphate removal efficiency varied between 40% and 84% in the aerobic setting and between 23% and 65% in the anaerobic setup, as depicted in Figure S3.

3.2. Continuous study

3.2.1. Temporal variation of organics

The reactor effluent COD was consistently low, with concentrations of 75–85 mg/L in phase 1 and 80–91 mg/L in phase 2, respectively. The results indicate significant efficiency in the flocculation or granulation of wastewater colloidal substances, promising the reactor's optimal functionality regarding organic compound oxidation and biomass proliferation. Despite the fact that a stabilised concentration was achieved at each stage, the average effluent COD consistently decreased throughout all phases. In phase 2, effluent COD reached levels anticipated for this high-efficiency treatment modality with a COD removal rate of 91%, as illustrated in Figures 1 and 2. A previous study has shown the removal of dissolved methane via an air stripper [12]. This study evaluated the potential outcomes for dissolved methane within the UASB, encompassing degasification via off-gas from the media sludge, stripping to the gas phase under mixing or scouring conditions, and biological oxidation by methane-oxidising bacteria (MOB) or methanotrophs. The effluent dissolved methane concentration ranged from 34 to 51 mg/L (±0.9) throughout this investigation. Methane was undetected in the headspace examination of aqueous UASB

effluent samples, but was recovered from the upper portion of the media in the UASB reactor, likely due to the oxidation of CH_4 by methanotrophic organisms or organic materials by heterotrophic bacteria. The supplementary aerobic heterotrophic degradation of soluble organic molecules was assessed by measuring the COD of UASB influent and effluent samples. UASB effluent samples were filtered using Whatman filter paper to quantify soluble COD following the elimination of solids generated by microbial action e.g. detached bio media; EPS (extracellular polymeric substances). The influent COD of the UASB was 310–500 (±10) mg/L, whereas the effluent COD was 80–91 mg/L, demonstrating a COD removal efficiency of 91% by the UASB system. The UASB effluent exhibited a TSS content of less than 128–129 mg/L in phases 1 and 2, thereby complying with the secondary effluent quality standards established by the USEPA, with a removal efficiency of 92%, as illustrated in Figure 1. The COD for the oxidation of sulphides and dissolved CH_4 was each less than 91 mg/L, with a removal efficiency of 91%, as illustrated in Figures S4–S5. The observed BOD values during the continuous study were lesser for nitrification (~200 mg/L), suggesting that CH_4 and sulphide-oxidising bacteria exceed for nitrifiers in oxygen utilisation, hence showing that the activity of nitrifiers was probably controlled by oxygen availability.

3.2.2. Temporal variation of sulphates and sulphides

The UASB influent SO_4^{2-} and H_2S concentrations was less than 85 mg/L (±1) before micro-aeration (Figure 3). After treatment with media with intermittent and continuous aeration in phases 1 and 2, the effluent SO_4^{2-} concentrations and removal effectiveness were not affected by variations of DO and were observed as 20–45 mg/L. The removal efficiency of sulphate was 65–70%. The SO_4^{2-} was completely separated by oxidation as it diffused into biofilms mounting on media installed in a reactor. This resulted in the swift elimination of sulphate and sulphide after the micro-aeration during phase 1 [condition 1–5] when the H_2S loading rate was less than 2.0 mg/L. During phase 2, when continuous aeration was given with recycling in the anaerobic zone and mixing (Figure 4), the sulphate decreased from 85 to 10 mg/L and gets further lowered due to decreased mixing between dissolved SO_4^{2-} and sulphur-oxidising bacteria (SOB).

The H_2S was detected in the UASB effluent at concentrations below 2 mg/L during phase 1. This data suggests that SOB eliminated sulphide within the plastic carrier situated in the anaerobic zone. Sulphate constituted the predominant sulphide oxidation product across all stages, representing over 90% of the influent sulphide, and its effluent removal diminished with continuous aeration. Particulate elemental sulphur raised concerns due to its contribution to effluent turbidity.

3.2.3. Temporal variation of NH_4-N

In contrast to the high sulphate removal efficiency talked about earlier, the UASB removed 85% of the ammonia in phase 2, and this number changed based on the operating conditions, as shown in Figure 4. The system was 75% effective at getting rid of ammonia when the ammonia concentration was 14 mg/L in phase 2 (condition 3–4). The removal efficiency dropped to 10% in phase 2 when the ammonia loading rate was 9 mg/L. This was because of the recycling and constant micro-aeration. In phase 2 [condition 3–4], the biofilm that formed on the media got thicker, which could mean that the nitrifiers stayed on the biofilm longer. When biofilm thickness was used as a media, the ammonia removal rate went up from 85% to 90% at the same ammonia influent concentration. The highest concentration of ammonia removed was 9 mg/L (Figure 3). Based on these results, it is possible that nitrifiers could grow on the outside of the biofilm. In Phase 2 [condition 4], the ability to remove ammonia was lowered by 86% to help a thicker biofilm grow by reducing turbulence. The NH_4^+-N removal efficiency decreased largely due to limited contact of NH_4^+ with nitrifiers [13]. In phase 2, continual micro-aeration under restricted mixing conditions, coupled with elevated nitrogen concentration in the influent, resulted in a further decline in efficiency. Ammonia in effluent and its oxidation products, such as nitrate was major products formed. The nitrate concentration in the influent was 2–3 mg/L prior to micro-aeration and increased to 4–6 mg/L in the effluent following micro-aeration. The efficacy of controlled ammonia removal varies from earlier lab-scale UASB research that accomplished simultaneous nitrification and anoxic sulphide oxidation. Both studies found that *Nitrospira* mostly oxidised ammonia to nitrate, then denitrified via the combination of sulphide oxidation with nitrate reduction enabled by *Thiobacillus*. Two-layer biofilms consisting of nitrifiers in the inner aerobic biofilm adjacent to the membranes with accessible oxygen and anoxic sulphide-oxidising bacteria on the outer biofilm where oxygen is limited, with nitrate reduction possibly producing sulphate or elemental sulphur, may be features of the lab-scale UASB. This work shows that aerobic respiration was preferred over nitrification in terms of sulphide removal, implying that aerobic SOBs dominated UASB biofilms and hence limited O_2 availability for nitrifiers. Nevertheless, it is noteworthy that variations in O_2 permeability

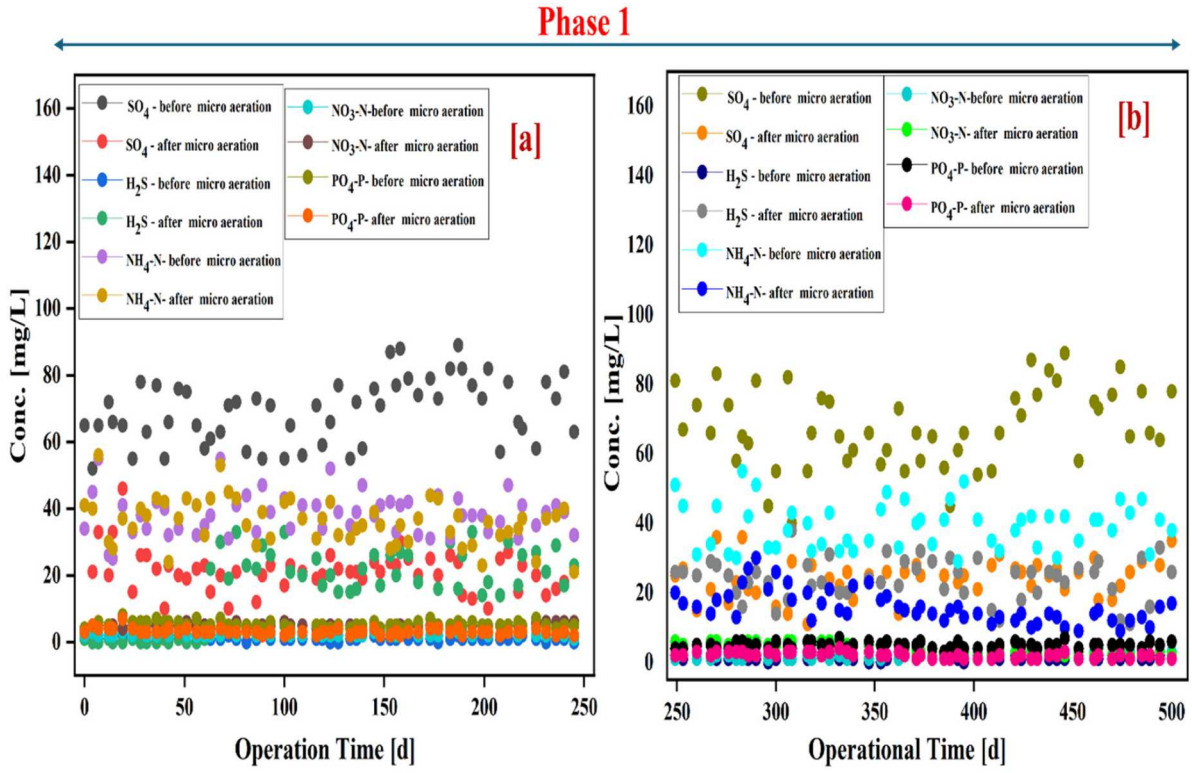


Figure 3. The concentration of SO_4, H_2S, NH_4-N, NO_3-N and PO_4-P from UASB reactor before and after micro-aeration [a] condition 1–3 and [b] condition 3–5 in phase 1.

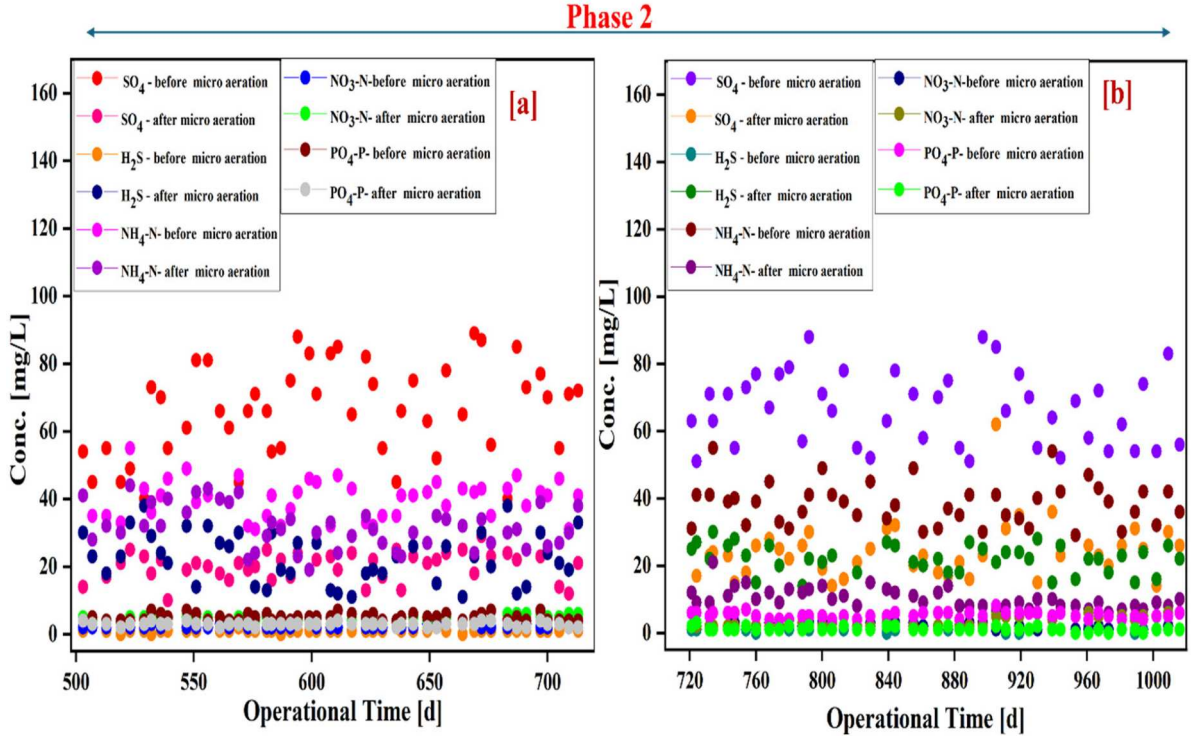


Figure 4. The concentration of SO_4, H_2S, NH_4-N, NO_3-N and PO_4-P from UASB reactor before and after micro-aeration [a] condition 1–2 and [b] condition 3–4 in phase 2.

and membrane surface properties could have influenced the membrane materials, therefore affecting the microbial community structure [14]. Previous laboratory-scale investigations employed polypropylene membranes or silicone tubing. This study differs from prior lab-scale investigations because of the more intricate influent utilised, which contained dissolved methane that may have modified the microbial ecology. The findings suggest that NH_4^+-N oxidation was constrained, although it could be partially regulated by the thickness of the biofilm. Partial nitrification in non-potable wastewater reuse applications can yield effluent that contains both nitrate (NO_3^-) and ammonia (NH_4^+) for use as fertiliser in irrigation. Though when underlying aquifers serve as clean water sources, the total residual-nitrogen concentration necessitates management to maintain adherence to the 9 mg/L Maximum Contaminant Level for NO_3^-. In clean wastewater recycle applications utilising RO, UASB can be employed to selectively oxidise sulphides, while ammonia can be discharged into the RO concentrate [15].

3.2.4. Temporal variation of PO_4-P

The UASB reactor with a microaerobic reaction phase was not intended for phosphorus removal. Consequently, in phase 1, phosphorus was predominantly eliminated to satisfy the metabolic requirements of heterotrophic species. In phase 2 setups, which include recycling in the anaerobic zone, the metabolic selection of slow-growing organisms, such as PAOs (poly-phosphate accumulating organisms) or DPAOs (denitrifying phosphate-accumulating organisms), was facilitated. PAOs are preferentially favoured at elevated P/COD ratios, whereas DPAOs exhibit enhanced development at reduced P/COD ratios [16]. Both PAOs and DPAOs assimilate COD under anaerobic circumstances, which benefit in energy conservation due to the lack of air supplies, in contrast to a completely aerobic grainy sludge reactor. Continuous microaerobic conditions were seen to effectively remove biologically available phosphorus in domestic wastewater, with levels exceeding 2 mg/L, which is approximately the minimum threshold necessary for phosphorus removal, as illustrated in Figure 4. As a result, the accumulation of PAOs was restricted, leading to the proliferation of DPAOs. It was anticipated that PAOs activity facilitated phosphorus removal in both phases, serving as the primary mechanism for phosphorus elimination throughout both stages of heterotrophic development. The involvement of other organisms liable for phosphorus removal, different from *Accumulibacter*, such as *Tetrasphaera* [17], which grows when the wastewater contains complex substrates. Beginning with phase 1, the phosphorus elimination attributable to PAO activity was minimal according to mass balances. The results indicate that PAOs exhibit significant sensitivity to operational conditions, as corroborated by recent studies from other authors [18]. It was the combination of a low P/COD ratio and micro-aeration feeding variations that inhibited PAOs.

3.3. FTIR characterisation and surface morphology

The chemical characterisation of anaerobic sludge at 0 days and final sludge at 1015 days from the reactors was conducted using FTIR spectroscopy (Figure 5). Enhanced bands of the aliphatic methylene groups at

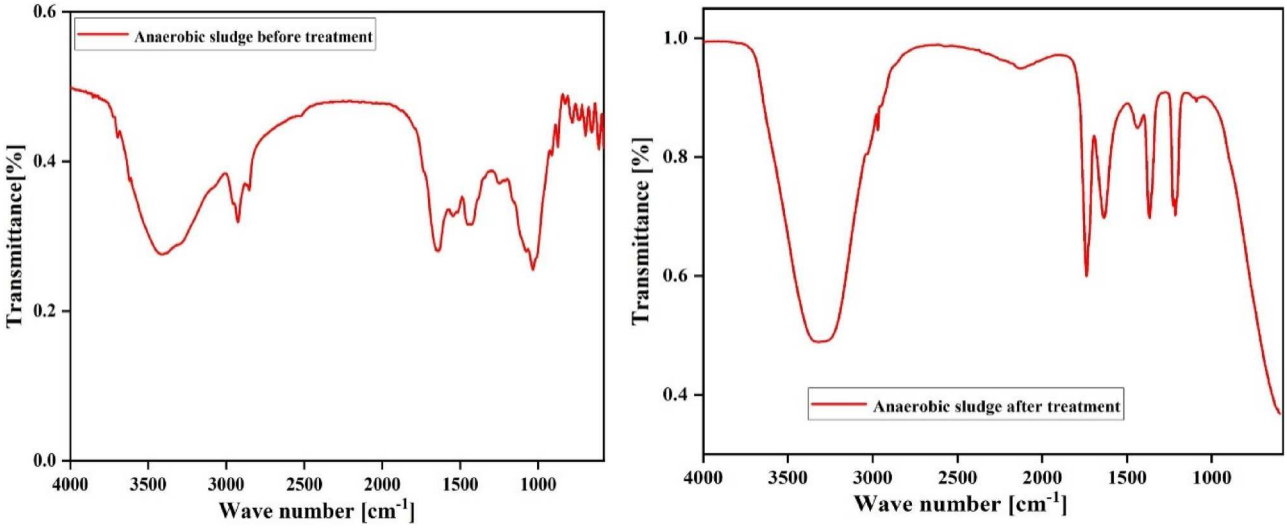


Figure 5. FTIR spectra of seed sludge before and after micro-aeration in UASB reactor.

2920 and 2854 cm^{-1}, in conjunction with distinct amide bands at 3478 cm^{-1}, were observed in anaerobic sludge prior to micro-aeration. Moreover, a pronounced O–H vibration of hydroxyl groups was also detected. A significant change in these bands was detected in seed sludge following micro-aeration. The three developing bands in the area of 890–1275 cm^{-1} in the spectra of the sludge indicate the existence of silica and carbohydrate compounds in the sludge [19]. The bands between 1646 and 1434 cm^{-1} in both sludge samples indicated the existence of bending vibrations of organic compounds, including phenolic O–H, N–H; (proteins and peptides), C=O vibrations of aldehyde; C=O stretching in amide groups, as well as carboxylic acids and primary amines [20]. The peak intensity of bands between 1500 and 1744 cm^{-1} in both sludge types increased following micro-aeration, indicating the presence of aromatic compounds and the advancement of the sludge.

Also, the absorbance peaks at 1355 and 1206 cm^{-1} demonstrated the non-appearance of inorganic functional groups [SO_4^{2-}] in the sludge post micro-aeration, confirming the occurrence of sulphur removal [21]. Figure S6 illustrates the FTIR spectrum of PVA gel, highlighting significant peaks corresponding to hydroxyl and acetate groups that have been identified. The significant bands observed at 3316 cm^{-1} correspond to the stretching of O–H bonds, occurring both intramolecularly and intermolecularly via hydrogen bonding. The vibrational band observed at 2915 cm^{-1} is associated with the stretching frequency of alkyl groups. A strong carbonyl group was identified, exhibiting a C=O stretching frequency between 1591 and 1640 cm^{-1}. The PVA gel bands in these locations are overlapping and expanding. Moreover, the O–H stretching vibration peak (3316 cm^{-1}) in the cross-linked PVA gel was significantly diminished relative to the pure PVA gel following the treatment conclusion. The ratio of the intensities of the vibrational bands of C=O (1591 cm^{-1}) and the bending vibration of CH_2 groups (1415 cm^{-1}) remained approximately constant. However, a notable alteration is evident in the bands at 1238, 1076, and 838 cm^{-1}, which may be ascribed to the absorption of organic compounds onto the PVA gel within the UASB reactor [22]. The FTIR spectra of the zeolite are presented in Figure S7 within the range of band vibrations (1220–580 cm^{-1}). A significant broad band was detected at 1009 cm^{-1}. Additional bands were detected at 875, 772, 766, 674, and 650 cm^{-1}. The bands result from the overlap of asymmetric vibrations of Si–O; (bridging) and Si–O– (non-bridging) bonds. The symmetric stretching of external links transpired at 1075 cm^{-1}, whereas the symmetric stretching resulting from the interior vibrations of the zeolite framework occurred at 674 cm^{-1}. The peak at 3285 cm^{-1} ascertained to the stretching vibration of the O-H; group of water adsorbed on the surface of zeolite. The predominant bands detected in zeolite following micro-aeration include a distinct absorption band at 1075 cm^{-1}, which corresponds to the asymmetric stretching vibrations of Si-O-Si [23]. The vibration band with peaks of 1739cm^{-1} is assigned to the bending vibration of absorbed water. The strong, wide band in the range 3285 cm^{-1} is associated with the presence of zeolite water. The bands at

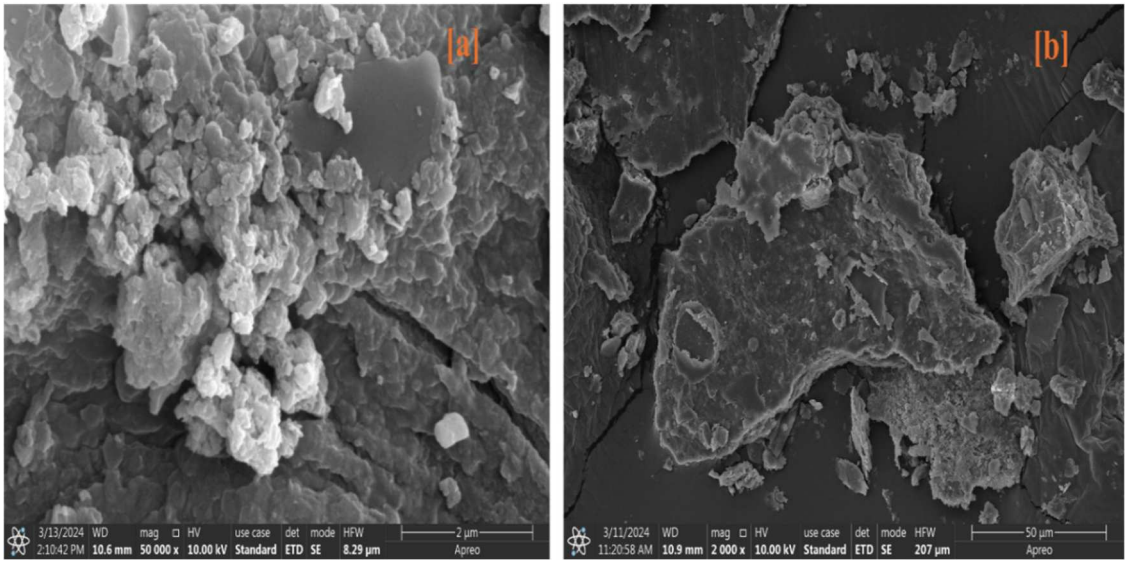


Figure 6. Scanning electron micrographs of sludge before and after micro-aeration. (a) Scanning electron micrographs of sludge before micro-aeration. (b) Scanning electron micrographs of sludge after micro-aeration.

3331 cm^{-1} arise due to asymmetric and symmetric stretching vibrations of isolated OH groups.

The structure and morphology of the biofilm in the sludge were examined using SEM, with the analogous images presented in Figure 6(a). The SEM studies indicated that the microbes in the sludge exhibited a compact arrangement with an irregular cauliflower morphology, characteristic of SOB (sulphur-oxidising bacteria) cultivation [24]. Simultaneously, the SEM images of sludge from the reactor exhibited a honeycomb; like arrangement (Figure 6(b)), further suggesting that these formations are mostly associated with coccoid bacterial cells, indicating a probable SOB culture. Nevertheless, these characteristics were absent in sludge samples at day zero, failing to confirm the existence of SOB bacteria. The SEM images of PVA gel were taken to examine the surface morphology (Figure 7(a,b)) and the cross-sectional morphologies of the PVA gel beads before and after treatment in the UASB reactor. The results demonstrated that the material exhibited a uniform and well-defined porosity in both its surface and cross section. Evidently, the pores on the surface exhibited heterogeneity in comparison to Figure 7(b). As the cross-linking reaction with microorganisms progressed, a dense layer formed on the surface of the PVA gel. This may have caused a decrease in the diffusion blend of the cross linking

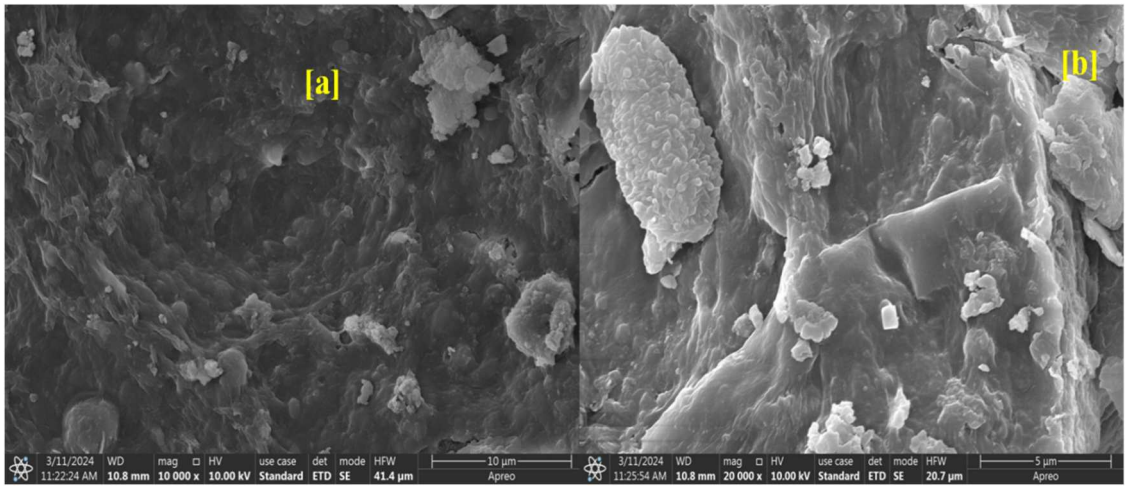


Figure 7. Scanning electron micrographs of PVA gel before and after micro-aeration. (a) Scanning electron micrographs of PVA gel before micro-aeration. (b) Scanning electron micrographs of PVA gel after micro-aeration.

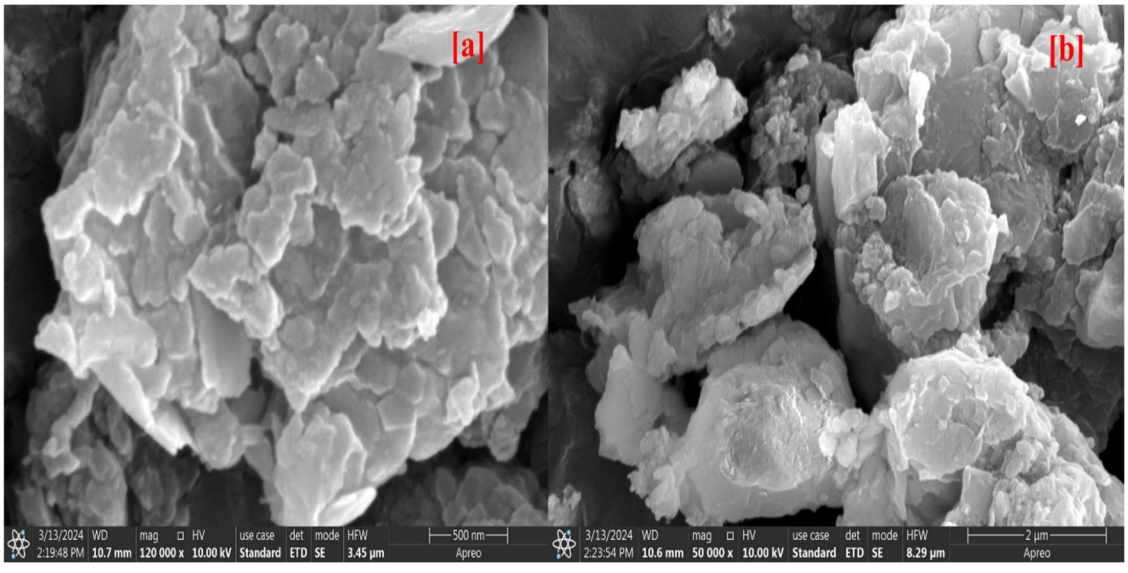


Figure 8. Scanning electron micrographs of zeolite before and after micro-aeration. (a) Scanning electron micrographs of zeolite before micro-aeration. (b) Scanning electron micrographs of zeolite after micro-aeration.

agent and slowed down the internal cross linking reaction. Consequently, the internal pores were reduced, and the gel became tightly packed. Put simply, the mismatch in the degree of cross linking inside and outside the material caused the core and shell to separate since they diffused at different speeds[25]. Figure 8(a,b) illustrates SEM images of the Zeolite sample before and after micro-aeration. The stick-shaped particles with size 10–18 µm and plate-shaped particles with size 2–6 µm are clearly observed. They aggregated into complex conglomerates called agglomerates. From 16a, we can see that the surface of native zeolite is smooth, wavy, and plate-like, with some pores. The structure inside is mostly uniform and thick. In Figure 8(b), it is easy to see that the zeolite biofilm has fine powdery particles sticking to the surface as well as thick plate-like holes that are caused by the growth of bacteria. This means that bacteria have been successfully squished onto the Zeolite's surface and some of its pores. O, Si, C, and Al are the main parts of zeolite. K is also present in small amounts. This, along with the plate-like structure seen in the scanning electron microscope, proves that zeolite is made of porous aluminosilicate. Microorganism-stuffed zeolite is mostly made up of O, Si, C, and Al. When bacterial growth was added to zeolite, the amount of oxygen in it went up, which means that microorganisms were properly attached to it [26].

4. Conclusions

The present study focused on the effect of micro-aeration for enhancing the removal of organics and nutrients with a short HRT 6 h of anaerobic process with DO levels ranging from less than 1.0 mg/L in phase 1 and between 1 and 1.5 mg/L in phase 2.

Significantly enhanced removal efficiency of organics and nutrients in the wastewater was observed. The removal of BOD, COD and TSS ranged from 80 to 90%, respectively at DO level <1.0 mg/L with continuous aeration. A batch study refers to the growth of the microbiota responsible for the removal of nitrogen. Extended exposure to microaerobic conditions over a period of 1015 days resulted in a progressive transformation of the microbiota. During all phases of study, sulphates were increased by more than 90% due to the sulphides oxidation after continuous aeration. The FTIR data suggested that organic molecules with phenolic O-H, N-H (proteins and peptides), and C=O vibration of aldehyde/carboxylic acid, respectively. The honeycomb-like structure shown in SEM pictures of reactor sludge further suggests that these structures are mostly associated with coccoid bacterial cells, most likely of NOBs and SOBs. Thus, micro-aerobic UASB reactors with integrated media have significant potential for scalable, efficient, and sustainable wastewater treatment, particularly in areas where energy, space, and operational simplicity are paramount. They signify a prospective advancement of anaerobic systems aimed at providing improved organic and nutrient removal with minimal added complexity.

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Author's contribution statement

Mahmood Ahmad: Writing – original draft. **Aaqib Ashraf Sheikh**: Software & Resources, Mohammad Ishteyaque Ahmad: Formal analysis and data Acquisitions. **Farooq Ahmad Wani**: Writing – review & editing. **Beni Lew**: Validation. **Abid Ali Khan**: Supervision, Resources

Disclosure statement

No potential conflict of interest was reported by the authors.

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Data availability statement

Any data and information used to support this research are present in the report.

References

[1] Grandclément C, Seyssiecq I, Piram A, et al. From the conventional biological wastewater treatment to hybrid processes, the evaluation of organic micropollutant removal: a review. Water Res. 2017;111:297–317. doi:10.1016/j.watres.2017.01.005

[2] Mert BK, Ozengin N, Dogan EC, et al. Efficient removal approach of micropollutants in wastewater using membrane bioreactor. In: Wastewater and Water Quality. London: IntechOpen; 2018. p. 53–63. doi:10.5772/intechopen.75183.

[3] Harb M, Zarei-Baygi A, Wang P, et al. Antibiotic transformation in an anaerobic membrane bioreactor linked to membrane biofilm microbial activity. Environ Res. 2021;200:111456. doi:10.1016/j.envres.2021.111456

[4] Ramadan L, Deeb R, Sawaya C, et al. Anaerobic membrane bioreactor-based treatment of poultry slaughterhouse wastewater: microbial community adaptation

and antibiotic resistance gene profiles. Biochem Eng J. 2023;192:108847. doi:10.1016/j.bej.2023.108847
[5] Vassalle L, Rueda E, Passos F, et al. Optimisation of high rate algal ponds performance for post-treatment of upflow anaerobic sludge blanket reactor effluents. Biosyst Eng. 2024;243:175–186. doi:10.1016/j.biosystemseng.2024.05.012
[6] Chernicharo C, Van Lier J, Noyola A, et al. Anaerobic sewage treatment: state of the art, constraints and challenges. Rev Environ Sci Bio/Technol. 2015;14:649–679. doi:10.1007/s11157-015-9377-3
[7] Krayzelova L, Bartacek J, Díaz I, et al. Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. Rev Environ Sci Bio/Technol. 2015;14:703–725. doi:10.1007/s11157-015-9386-2
[8] Siqueira JPS, Pereira AM, Dutra AMM, et al. Process bioengineering applied to BTEX degradation in microaerobic treatment systems. J Environ Manage. 2018;223:426–432. doi:10.1016/j.jenvman.2018.06.066
[9] Buarque PMC, RBPd L, Vidal CB, et al. Enhanced removal of emerging micropollutants by applying microaeration to an anaerobic reactor. Engenharia Sanitaria e Ambiental. 2019;24:667–673. doi:10.1590/s1413-41522019201920190030
[10] Nokman W, Benluvankar V, Packiam SM, et al. Screening and molecular identification of heavy metal resistant pseudomonas putida S4 in tannery effluent wastewater. Biocatal Agric Biotechnol. 2019;18:101052. doi:10.1016/j.bcab.2019.101052
[11] Gonzalez-Gil L, Fernandez-Fontaina E, Singh RR, et al. Feeding composition and sludge retention time both affect (co-)metabolic biotransformation of pharmaceutical compounds in activated sludge systems. J Environ Chem Eng. 2021;9:105123. doi:10.1016/j.jece.2021.105123
[12] Stackebrandt E. Taxonomic parameters revisited: tarnished gold standards. Microbial Today. 2006;33:152.
[13] Sengar A, Aziz A, Farooqi IH, et al. Development of denitrifying phosphate accumulating and anammox micro-organisms in anaerobic hybrid reactor for removal of nutrients from low strength domestic sewage. Bioresour Technol. 2018;267:149–157. doi:10.1016/j.biortech.2018.07.023
[14] Ashish Sengar AS, Asad Aziz AA, Farooqi I, et al. Development of denitrifying phosphate accumulating and anammox micro-organisms in Anaerobic Hybrid Reactor for removal of nutrients from low strength domestic sewage. Bioresour Technol. 2018;267:149–157. doi:10.1016/j.biortech.2018.07.023
[15] Gonzalez-Gil L, Carballa M, Lema JM. Removal of organic micro-pollutants by anaerobic microbes and enzymes. In: Current Developments in Biotechnology and Bioengineering. Amsterdam: Elsevier; 2020. p. 397–426.
[16] Weissbrodt DG, Shani N, Holliger C. Linking bacterial population dynamics and nutrient removal in the granular sludge biofilm ecosystem engineered for wastewater treatment. FEMS Microbiol Ecol. 2014;88:579–595. doi:10.1111/1574-6941.12326
[17] Martínez-Campos S, González-Pleiter M, Fernández-Piñas F, et al. Early and differential bacterial colonization on microplastics deployed into the effluents of wastewater treatment plants. Science of the Total Environment. 2021;757:143832. doi:10.1016/j.scitotenv.2020.143832
[18] Wang J-H, Zhang T-Y, Dao G-H, et al. Microalgae-based advanced municipal wastewater treatment for reuse in water bodies. Appl Microbiol Biotechnol. 2017;101:2659–2675. doi:10.1007/s00253-017-8184-x
[19] Stevenson FJ. Humus chemistry: genesis, composition, reactions. New York: John Wiley & Sons; 1994.
[20] Gigliotti G, Businelli D, Giusquiani PL. Composition changes of soil humus after massive application of urban waste compost: a comparison between FT-IR spectroscopy and humification parameters. Nutr Cycling Agroecosyst. 1999;55:23–28. doi:10.1023/A:1009829008018
[21] Liang Z, Liu J. Landfill leachate treatment with a novel process: anaerobic ammonium oxidation (Anammox) combined with soil infiltration system. J Hazard Mat. 2008;151:202–212. doi:10.1016/j.jhazmat.2007.05.068
[22] Mansur HS, Sadahira CM, Souza AN, et al. FTIR spectroscopy characterization of poly (vinyl alcohol) hydrogel with different hydrolysis degree and chemically crosslinked with glutaraldehyde. Mater Sci Eng C. 2008;28:539–548. doi:10.1016/j.msec.2007.10.088
[23] Breck DW. Zeolite molecular sieves: structure, chemistry, and use. New York: John Wiley & Sons; 1974.
[24] Trigo C, Campos J, Garrido J, et al. Start-up of the Anammox process in a membrane bioreactor. J Biotechnol. 2006;126:475–487. doi:10.1016/j.jbiotec.2006.05.008
[25] Sun L, Wang J, Liang J, et al. Boric acid cross-linked 3D polyvinyl alcohol gel beads by NaOH-titration method as a suitable biomass immobilization matrix. J Polym Environ. 2020;28:532–541. doi:10.1007/s10924-019-01610-z
[26] Shi Y, Wang X, Feng C, et al. Adsorption properties of natural zeolite supported by chitosan on Cd (II) in micropolluted irrigation water. ACS Omega. 2023;9:573–584.